



Optimizing temperature, light intensity, and salinity for carpospore culture of *Gracilaria fisheri*

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ABSTRACT

Gracilaria is a genus of economically important macroalgae that is widely cultivated worldwide. Currently, *Gracilaria* production relies on vegetative propagation, which is a labor-intensive method that leads to germ-plasm degradation and reduced yield. Spore-based seedling cultivation offers an effective solution to these issues. In this study, we conducted a series of orthogonal experiments to investigate the optimal temperature, light intensity, and salinity conditions for cultivation of *Gracilaria fisheri* carpospores. Through these experiments, we systematically explored the entire cultivation process to elucidate the variations in carpospore release quantity, attachment rate, and germination rate and in the carposporeling growth rate and pigment content under different combinations of conditions. We found that the optimal conditions for carpospore release were 25°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity. For carpospore attachment, the best conditions were 20°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity. The ideal conditions for carpospore germination were 30°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity. The most favorable conditions for carpospore seedling growth were 25°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity, and these temperature and salinity conditions were also favorable for pigment accumulation. However, chlorophyll *a* and carotenoids accumulation was higher under the lower light intensity condition (20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$). Under suitable conditions, visible carpospores of *G. fisheri* were observed within 4 weeks, and the maximum length reached 0.79 cm after 9 weeks. These findings provide a theoretical foundation and practical guidance for the cultivation, genetic breeding, and artificial spore propagation of *Gracilaria* species.

1. Introduction

With over 7000 recognized species, red algae constitute a rich and distinctive group of seaweeds (Guiry, 2024). Humans have consumed red algae for thousands of years (Collen et al., 2014; Tseng et al., 1962), and they are increasingly valued as a healthy, sustainable food source (Rawiwan et al., 2022). Rich in polysaccharides, derivatives, minerals, and phenolic lipid compounds, they are well-suited for carrageenan and agar extraction, as well as applications in biotechnology and the pharmaceutical sector (Cian et al., 2015; Cole and Sheath, 1990; Dhana-lakshmi and Jayakumari, 2018). In aquaculture and environmental restoration, red algae serve as biofilters because they absorb nitrogen, phosphorus, and heavy metals from the water, thereby enhancing water quality (Hashim and Chu, 2004; Schuenhoff et al., 2006). Through allelopathic effects, red algae also inhibit the growth of harmful bacteria

and algae, which supports the health of cultivated species (Huang et al., 2022; Patil et al., 2020). These features position red algae as an essential element in multi-trophic aquaculture systems (Ashkenazi et al., 2019; Felaco et al., 2020).

Gracilaria is a significant macroalgal genus within the phylum Rhodophyta (red algae), class Florideophyceae, order Gracilariales, and family Gracilariaceae, and the genus contains over 200 documented species (Guiry and Guiry, 2024). These macroalgae predominantly thrive in temperate, subtropical, and tropical regions (Bird and McLachlan, 1984; Sijian, 1987), with the highest species diversity found in the tropical waters of the Indian Ocean, Pacific Ocean, and western Atlantic Ocean (de Oliveira, 1984; Guiry, 2024; Gurgel and Fredericq, 2004). As industrial progress drove a rising demand for agar in the early twentieth century, the limited production capacity of the red alga *Gelidium* could not meet market needs, which enhanced the importance of

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Gracilaria in agar production. By the late twentieth century, the cultivation of *Gracilaria* had grown significantly in developing nations such as China, Indonesia, and Chile, which emerged as leading global producers (Armisen, 1995; McHugh, 1991).

At present, *Gracilaria* cultivation predominantly relies on vegetative propagation, whereby thallus fragments are used for reproduction (Bezerra and Marinho-Soriano, 2010; Ryder et al., 2004). However, this method demands significant labor and requires that part of the biomass be reserved as starter material, which prevents complete harvesting of the thallus (Glenn et al., 1996; Mantri et al., 2009). Hurtado-Ponce et al., (1992) reported that *Gracilaria* farmers in the Philippines allocate up to 30 % of their harvest as initial propagation material for subsequent cycles. Furthermore, repeated vegetative propagation can result in diminished growth rates and yields in subsequent generations, likely due to aging or germplasm degradation (Buschmann et al., 1995; Oliveira et al., 2000).

To address these limitations, the large-scale *Gracilaria* production industry requires alternative propagation methods, such as sexual reproduction. Similar to other Florideophyceae algae, *Gracilaria* has a life cycle consists of three generations (diploid tetrasporophytes and carposporophytes and haploid male and female gametophytes), and they exhibit isomorphic alternation of generations (Kain and Destombe, 1995). The utilization of carpospores or tetraspores as seed stock can reduce the initial material needed for *Gracilaria* cultivation (Halling et al., 2005). Mantri et al. (2009) reported that carpospore seedling production via sexual reproduction offers a sustainable supply of uniform seedlings. Additionally, carposporelings, which are the early-stage tetrasporophytes that germinate from carpospores, demonstrate robust adaptability to environmental conditions, which makes them outstanding candidates for cultivation.

Numerous studies have focused on spore reproduction in *Gracilaria*. For example, Yu et al. (2013) reported that the optimal salinity for carpospore growth and development in *G. edulis* and *G. tenuistipitata* is 15, with a favorable light intensity range of 60–130 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$. They also identified the optimal salinity for carpospore germination of these species as 28–30, with an ideal temperature range of 23–27°C. Xie et al. (2010) found that the PSII photosynthetic capacity of *G. vermiculophylla* tetraspores increased significantly following successful attachment, with photosynthetic activity becoming uniform across all cells after the formation of multicellular disks. Guzmán-Urístegui and Robledo (1999) noted that *G. cornea* carpospores had peak dispersal within the first 3 days, with maximum release rates observed at 26°C. Mantri et al. (2009) further reported a daily release rate of 808 carpospores per cystocarp in *G. dura*, with tetraspores exhibiting a daily growth rate of approximately 4.67 % d^{-1} and reaching lengths of 3–4 cm after 6 months. While these studies were focused primarily on specific stages of carpospore reproduction, investigating the optimal conditions for each phase of the complete reproductive cycle is vital to further reduce the seedling cultivation time of *Gracilaria*.

Gracilaria fisheri is native to the coastal regions of Southeast Asia and the South China Sea (Bangmei and Abbott, 1987; Phang et al., 2016). This species is commercially cultivated in Asia primarily for consumption and agar extraction, but bioactive compounds in the thalli also have medicinal potential (Benjama and Masniyom, 2012; Nunraksa et al., 2019). *G. fisheri* is also capable of absorbing nutrients and heavy metals from the environment, making it a promising species for ecological aquaculture (Chaisuksant, 2003; Kaewsuralikhit, 1994). While studies have been conducted on tissue culture and spore propagation of *G. fisheri* (Phi, 2014; Sangtong et al., 2020), research focusing on the optimization of carpospore cultivation conditions remains scarce. Therefore, we used an orthogonal experimental approach to study how temperature, light intensity, and salinity influence the release, attachment, germination, and seedling growth of *G. fisheri* carpospores. The objectives of this study were to identify the optimal conditions for each stage of carpospore cultivation to expedite the completion of its life cycle and to provide theoretical and practical guidance for seedling proliferation and

large-scale aquaculture under controlled conditions.

2. Materials and methods

2.1. Experimental materials and pretreatment

The *G. fisheri* samples used in this study were collected in March 2024 from Wushi Town, Leizhou City, Zhanjiang, Guangdong Province, China (109°50'9.60"E, 20°33'18.00"N). After transport to the Algae Laboratory at Guangdong Ocean University, the thalli were rinsed with sterile seawater. Female gametophytes bearing cystocarps were selected and immersed in a 1 % povidone-iodine solution for 1 min to remove debris and fouling organisms (Borburema et al., 2021). The treated samples were then transferred to sterile seawater with 25–30 salinity and kept in an illuminated incubator for 3 days under controlled conditions: 25°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and a 12 light:12 dark. These *G. fisheri* samples were then used for subsequent experiments.

2.2. Experimental design and data processing

To minimize experimental costs and determine the optimal conditions for each stage of *G. fisheri* carpospore reproduction, we employed the orthogonal design method. The experiment to determine the best conditions for carpospore release, attachment, and germination included three factors (temperature, light intensity, and salinity), and each had four levels. An orthogonal three factor four level (L_4^3) array was used to set up 16 experimental groups, each with three replicates (Table 1). For the experiment to study carpospore seedling growth, the same three factors and four levels were employed. Due to limited experimental materials, an L_4^4 orthogonal array was used to design 16 groups without replication, with one blank column reserved as an error column. The factors and levels used in both orthogonal experiments were identical (Table 1). The selection of factor levels was based on appropriate ranges reported in existing literature (Ashok et al., 2016; Chirapart and Ohno, 1993; Lewmanomont and Kaewsuralikhit, 1993; Nguyen et al., 2017; Yu et al., 2013).

The results of the 16 combinations were evaluated using range analysis and one-way analysis of variance. The mean values for levels one to four of each factor were represented as k1, k2, k3, and k4, respectively. The optimal condition for each factor was determined by the maximum mean value (k_{max}) (Cornejo-Mazon et al., 2008; Venkata and Chandra, 2008). The range (R), which represents the difference between the maximum and minimum mean values for each factor, was used to gauge variation and rank the factors. Analysis of variance was conducted to assess the contribution and significance of each factor on the outcome. When a blank column was present in the orthogonal design, it was considered to be an experimental error.

Statistical analyses were performed using SPSS 20 (IBM, Armonk, NY, USA), with significance set at $P < 0.05$. Data for carpospore release were expressed as mean $\pm$ standard error.

2.3. Optimization of conditions for carpospore release

The optimization experiment for *G. fisheri* carpospore release conditions was conducted over a period of 7 days. Sixteen treatment groups were established according to an orthogonal design, with each group

Table 1
Levels and factors of the orthogonal experimental design.

Factor	A Temperature (°C)	B Light intensity ($\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$)	C Salinity
Level1	15	20	20
Level2	20	40	25
Level3	25	80	30
Level4	30	160	35

consisting of three independent replicates. *G. fisheri* samples were randomly selected, and cystocarp-bearing branches were excised using a sterile scalpel. Excess moisture was blotted away with sterile cotton, and segments with evenly distributed cystocarps (approximately 3–5 cm in length) were selected. To prepare the media, sterile seawater was adjusted to varying salinities by adding distilled water or artificial sea salts, and Von Stosch Enrichment (VSE) medium was prepared according to the Von Stosch formula (von Stosch, 1964) using sterile seawater and GeO_2 to produce VSE seawater medium (Yarish et al., 2012). For each replicate, a 0.3 g *G. fisheri* segment was placed in a 90-mm Petri dish containing 30 mL of VSE seawater medium. The Petri dishes were placed in an incubator with conditions controlled as per the orthogonal design.

Each day, carpospore counts per milliliter were recorded using an upright microscope (Olympus BX53, Tokyo, Japan) and a plankton counting chamber. The medium and Petri dishes were refreshed daily. Prior to transferring segments to new dishes, they were rinsed in fresh VSE medium. After each handling, samples were promptly returned to their designated culture environment. The mean daily concentrations of spores over the 7-day period were calculated based on the observed carpospore concentrations and served as an indicator of carpospore production.

2.4. Optimization of conditions for carpospore attachment and germination

In the experiment to optimize attachment and germination conditions for *G. fisheri* carpospores was conducted over a 30-day period. *G. fisheri* specimens with cystocarps were placed in beakers containing 1000 mL of VSE seawater medium, and the number of carpospores in the medium was recorded daily. When substantial carpospore release occurred, the *G. fisheri* samples were removed, and the medium was filtered through an 80-mesh sieve. The filtrate was adjusted to a specific salinity, creating spore water with an approximate density of 400 ind/mL.

In a 12-well plate, three central wells (B2, B3, and C2) were selected, and a 1 cm² rectangle was etched onto the underside of each well to designate the counting area. Following the conditions established by the orthogonal design, 2 mL of spore water at the specified salinity were added to each well. Plates were then placed in incubators set to the respective light and temperature conditions and cultured for 30 days. At the start of the experiment, the number of carpospores within the designated area was counted using an inverted microscope (Zeiss Axio Vert.A1, Oberkochen, Germany). Subsequently, the culture medium in the wells was refreshed every 7 days using a pipette. Every 5 days, the number of carpospores, disks, and erect thalli in the area was recorded. An upright microscope (Olympus BX53, Tokyo, Japan) was used during the experiment to monitor the developmental process of carpospores. Key time points for carpospore attachment, germination, and morphological changes were recorded. The diameters of carpospores and disks, along with carposporeling lengths at each stage, were also measured.

The attachment rate and germination rate were calculated as follows (Garza-Sánchez et al., 2000; Marchi and Plastino, 2020):

$$R_A = N_{A_{\max}}/N_S \times 100\%$$

$$R_G = N_{G_{\max}}/N_S \times 100\%$$

where R_A is the attachment rate, $N_{A_{\max}}$ is the maximum count of multicellular disks within the area over 30 days, N_S is the initial spore count on day 0, R_G is the germination rate, and $N_{G_{\max}}$ is the maximum count of erect thalli over 30 days.

2.5. Optimization of growth conditions for carposporelings

After maintaining the carposporelings from the previous experiment

under acclimation conditions, growth continued for 1 month until they reached a visible size. These visible carposporelings were collected and used in the subsequent experiments. Sixteen treatment groups were set up following an orthogonal design, with no replication due to limited carposporeling numbers. Sixteen Petri dishes (90 mm in diameter) were prepared, each containing VSE seawater medium at a specific salinity. Using sterile tweezers, 40 carposporelings were inoculated into the medium in each dish. The Petri dishes were then placed in a light-controlled growth chamber and cultured under the specified conditions for 35 days. Initial carposporeling lengths were measured using a stereoscopic fluorescence microscope (Nikon Z18, Tokyo, Japan), and the culture medium was replaced every 7 days. The linear growth rate of the carposporelings was calculated as follows:

$$V(\text{mm/day}) = (L - L_0)/T$$

where V is the linear growth rate, L is the final carposporeling length at the end of the experiment, L_0 is the initial length, and T is the number of culture days.

After 35 days, the photosynthetic pigment content of carposporelings was measured for each treatment group. Chlorophyll *a* (Chl_a) and carotenoids (Car) were extracted using a methanol extraction method. A 10 mg sample of algae was placed in a grinding tube with four grinding beads and 2 mL of methanol. Samples were processed in a tissue grinder (JFstgrp-24, Shanghai, China) at 70 Hz for 180 s and then stored in the dark at 4°C for 24 h. The extracts were centrifuged (Cence H1650-W, Hunan, China) at 1957 × g for 10 min, and the absorbance of the supernatant was measured at 480, 510, 652, 665, and 750 nm using a UV spectrophotometer (Shimadzu UV-2450, Kyoto, Japan). Chl_a and Car contents were calculated as follows (Lichtenthaler, 1987; Porra et al., 1989):

$$W_{\text{Chl}_a} = [16.29 \times (A_{665} - A_{750}) - 8.54 \times (A_{652} - A_{750})] \times V / (1000 \times F_W)$$

$$W_{\text{Car}} = \{7.6 \times [(A_{480} - A_{750}) - 1.49 \times (A_{510} - A_{750})]\} \times V / (1000 \times F_W)$$

where W_{Chl_a} and W_{Car} are the mass fractions of Chl_a and Car (mg·g⁻¹), respectively, A is the absorbance, V is the volume of pigment extraction solution (mL), and F_W is the fresh weight of the sample (g).

For the phycobiliproteins phycoerythrin (PE) and phycocyanin (PC), pigments were extracted using 0.1 mol·L⁻¹ phosphate buffer (pH 6.8) using the same method as for Chl_a and Car. PE and PC contents were calculated as follows (Beer and Eshel, 1985):

$$W_{\text{PE}} = [(A_{564} - A_{592}) - (A_{455} - A_{592}) \times 0.20] \times 0.12 \times V / F_W$$

$$W_{\text{PC}} = [(A_{618} - A_{645}) - (A_{592} - A_{645}) \times 0.51] \times 0.15 \times V / F_W$$

where W_{PE} and W_{PC} are the mass fractions of PE and PC (mg·g⁻¹), respectively, A is the absorbance, V is the final solution volume (mL), and F_W is the fresh weight of the algal sample (g).

3. Results

3.1. Optimal conditions for carpospore release

Carpospore release was detected within 6–12 h of the experiment's initiation and persisted throughout the 7-day study period. Substantial differences in carpospore release rates were observed among the treatment groups (Table 2). Treatment group 16 exhibited the highest mean daily concentrations of spores at 189 ind mL⁻¹ under conditions of 30°C, 160 μmol·m⁻²·s⁻¹, and 30 salinity and was followed closely by group 13, at 188 ind mL⁻¹ under 30°C, 20 μmol·m⁻²·s⁻¹, and 35 salinity. Conversely, the lowest mean daily concentrations of spores (3 ind mL⁻¹) was observed in group 2 under conditions of 15°C, 40 μmol·m⁻²·s⁻¹, and 25 salinity.

Carpospore release initially increased and then decreased with rising

Table 2
Orthogonal experiment and calculated results of carpospore release.

Group	Factor			The mean daily concentrations of spores (ind/mL)
	A	B	C	
1	15	20	20	5 ± 2
2	15	40	25	3 ± 0
3	15	80	30	11 ± 1
4	15	160	35	3 ± 1
5	20	20	25	4 ± 0
6	20	40	30	24 ± 4
7	20	80	35	12 ± 3
8	20	160	20	3 ± 1
9	25	20	30	123 ± 17
10	25	40	35	142 ± 36
11	25	80	20	121 ± 16
12	25	160	25	106 ± 4
13	30	20	35	188 ± 23
14	30	40	20	30 ± 3
15	30	80	25	67 ± 12
16	30	160	30	189 ± 25
The range analysis of orthogonal experiment results for carpospore release				
k1	5	80	40	
k2	11	50	45	
k3	123	53	87	
k4	119	75	86	
R	118	30	47	

k1, k2, k3 and k4 were the average values of Level 1, Level 2, Level 3 and Level 4 for each factor.
The range (R) was the difference between the maximum and minimum average ($k_{\max}-k_{\min}$) for each factor.
 $k_{\max}$ and $k_{\min}$ were the maximum and the minimum value of k1, k2, k3 and k4, respectively.

temperature, reaching its maximum ($k_3 = 123$) at 25°C (Table 2). The mean daily concentrations of spores at 25°C was approximately 24 times higher than that at 15°C ($k_1 = 5$). Similarly, carpospore release increased with salinity, with the highest mean daily concentrations of spores

peaking at 30 ($k_3 = 87$). Variance analysis indicated that temperature and salinity significantly influenced carpospore release ($df = 3, N = 48, F = 37.98, P < 0.01$; $df = 3, N = 48, F = 5.80, P < 0.01$), while light intensity did not have a significant effect ($df = 3, N = 48, F = 37.98, P = 0.11$). Based on the magnitude of variation of carpospore release, the factors affecting the average daily release were ranked as follows: temperature ($R = 118$) > salinity ($R = 47$) > light intensity ($R = 30$) (Table 2), which is consistent with the results of the variance analysis. Based on the $k_{\max}$ values (Table 2), the optimal conditions for carpospore release were identified as 25°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity.

3.2. Optimal conditions for carpospore attachment and germination

Table 3 present the mean attachment and germination rates of carpospores for each treatment group after 30 days of cultivation. Under conditions of 25°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity, group 9 achieved the highest attachment rate (67.8 %) (Table 3). In contrast, no multicellular disks were observed in treatment groups 1, 2, 3, 4, 8, and 14, resulting in attachment rates of 0 % for these groups. The attachment rate of carpospores increased initially with rising salinity, peaking at 30 ($k_3 = 44.7$ %), but decreased sharply to only 0.93 % at 20 salinity. A similar pattern was observed for temperature and light intensity: attachment rates peaked at 20°C ($k_2 = 33.6$ %) and 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ ($k_2 = 29.2$ %) before declining. At 15°C, no multicellular disks formed, resulting in an attachment rate of 0 %. Variance analysis demonstrated that temperature, light intensity, and salinity significantly affected attachment rates ($df = 3, N = 48, F = 22.09, P < 0.01$; $df = 3, N = 48, F = 5.07, P = 0.04$; $df = 3, N = 48, F = 31.52, P < 0.01$), and the order of influence was as follows: salinity ($R = 43.8$ %) > temperature ($R = 33.6$ %) > light intensity ($R = 16.6$ %), which is consistent with the variance analysis results.

Cultivated at 30°C, 160 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity, treatment group 16 exhibited the highest germination rate (38.4 %), and it was followed closely followed by treatment group 9 under conditions of

Table 3
Orthogonal experiment and calculated results of attachment rate and germination rate.

Group	Factor			Attachment rate	Germination rate
	A	B	C		
1	15	20	20	0.0 %	0.0 %
2	15	40	25	0.0 %	0.0 %
3	15	80	30	0.0 %	0.0 %
4	15	160	35	0.0 %	0.0 %
5	20	20	25	28.7 %	0.0 %
6	20	40	30	60.8 %	33.9 %
7	20	80	35	44.8 %	0.0 %
8	20	160	20	0.0 %	0.0 %
9	25	20	30	67.8 %	38.3 %
10	25	40	35	56.2 %	21.2 %
11	25	80	20	3.8 %	3.8 %
12	25	160	25	0.2 %	0.2 %
13	30	20	35	12.3 %	7.0 %
14	30	40	20	0.0 %	0.0 %
15	30	80	25	35.8 %	30.0 %
16	30	160	30	50.4 %	38.4 %
The range analysis of orthogonal experiment results for Attachment rate					
k1	0.0 %	27.2 %	1.0 %		
k2	33.6 %	29.2 %	16.9 %		
k3	32.0 %	21.1 %	44.7 %		
k4	24.6 %	12.7 %	28.3 %		
R	33.6 %	16.6 %	43.8 %		
The range analysis of orthogonal experiment results for Germination rate					
k1	0.0 %	11.3 %	1.0 %		
k2	8.5 %	13.8 %	7.6 %		
k3	15.9 %	8.5 %	27.6 %		
k4	18.8 %	9.7 %	7.1 %		
R	18.8 %	5.3 %	26.7 %		

k1, k2, k3 and k4 were the average values of Level 1, Level 2, Level 3 and Level 4 for each factor.
The range (R) was the difference between the maximum and minimum average ($k_{\max}-k_{\min}$) for each factor.
 $k_{\max}$ and $k_{\min}$ were the maximum and the minimum value of k1, k2, k3 and k4, respectively.

25°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity (38.3 %) (Table 3). No multicellular disk was observed in treatment groups 1, 2, 3, 4, 8, and 14, resulting in germination rates of 0 %. While multicellular disks were present in treatment groups 5 and 7, none successfully germinated into erect thalli, leading to germination rates of 0 %. The germination rate of carpospores increased initially with rising salinity, peaking at 30 ($k_3 = 27.6$ %) before declining. The germination rate of carpospores increased with rising temperature, where germination rates progressively increased, reaching a maximum at 35°C ($k_4 = 18.8$ %). Variance analysis indicated that both salinity and temperature significantly influenced carpospore germination ($df = 3$, $N = 48$, $F = 11.68$, $P < 0.01$; $df = 3$, $N = 48$, $F = 6.14$, $P < 0.01$). However, light intensity had no significant effect, because its MS value (0.01) was smaller than the MSe (0.02) and thus was combined into the error term. Among these factors, the order of influence on germination rates was as follows: salinity ($R = 26.7$ %) > temperature ($R = 18.8$ %) > light intensity ($R = 5.3$ %), which is consistent with the variance analysis results.

Overall, salinity and temperature significantly influenced both the attachment and germination rates of carpospores, while light intensity only had a significant effect on attachment rates. Based on the k_{max} values shown in Table 3, the optimal conditions for carpospore attachment were 20°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity, while the optimal conditions for carpospore germination were 30°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity.

3.3. Morphological observations of carpospore development

G. fisheri carpospores were spherical, measuring between 13 and 22 μm in diameter, with densely packed red pigments (Fig. 1a). After being discharged from the cystocarp, the carpospores floated in the water (Fig. 2a), and under optimal conditions they naturally settled to the container's bottom within 2 days, where they began cell division and attachment (Fig. 1b). A layer of adhesive material surrounded newly released carpospores, allowing them to adhere to the container's surfaces. When large quantities of carpospores accumulated on the container bottom, they formed a visible pinkish sediment (Fig. 2b).

During the first week of the experiment, carpospore yield reached up to 18,900 per gram, with approximately 280 carpospores released daily from each cystocarp. The first division occurred 12–24 h after release and divided the cell into two equal parts along a central plane (Fig. 1b). The second division followed a perpendicular axis to the first, with no increase in cell volume (Fig. 1c). Between days 2 and 5, multiple rounds of division increased both cell count and the overall size of the structure (Fig. 1d, e). A multicellular disk formed within 5–7 days under suitable conditions (Fig. 1f). The multicellular disk diameter ranged from 28 to 56 μm . The formation of the disk signified successful attachment of the carpospores to the container wall. Cells in the center of the disk then

began dividing, elevating, and extending several transparent filaments outward (Fig. 1g). After approximately 2 weeks, an erect thallus emerged from the center of the disk (Fig. 1h). Within 4–5 weeks, the erect thallus transitioned into a visually detectable carposporeling, which reached a length of 0.13–0.79 cm by week 9.

Furthermore, we observed that carpospores failing to attach promptly after release ruptured and eventually died (Fig. 2c). Adjacent disk occasionally fused during development (Fig. 2d). The color of carpospore seedlings varied under different growth conditions (Figs. 2e, 2f).

3.4. Optimal growth conditions for carposporelings

Table 4 shows the linear growth rates of carpospore seedlings after 35 days of cultivation. Treatment group 14 exhibited the highest linear growth rate of 0.07 $\text{mm}\cdot\text{day}^{-1}$ under conditions of 30°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 35 salinity. In contrast, treatment groups 1, 2, 4, 5 had linear growth rates below 0.02 $\text{mm}\cdot\text{day}^{-1}$, with group 1 showing negative growth under conditions of 15°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 20 salinity. The linear growth rate of carpospore seedlings increased initially and then decreased with rising temperature, light intensity, and salinity. Variance analysis revealed that temperature and salinity had significant impacts on the growth rate of carpospore seedlings ($df = 3$, $N = 16$, $F = 55.80$, $P < 0.01$; $df = 3$, $N = 16$, $F = 35.70$, $P < 0.01$), while light intensity had no significant effect ($df = 3$, $N = 16$, $F = 4.58$, $P = 0.12$). Based on the magnitude of growth rate variation (Table 4), the factors influencing growth rates were ranked as follows: temperature ($R = 0.04$) > salinity ($R = 0.03$) > light intensity ($R = 0.01$), which is consistent with the results of variance analysis. Based on the k_{max} values (Table 4), the optimal conditions for linear growth rate of carposporelings were identified as 25°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity.

3.5. Influence of temperature, light intensity, and salinity on pigment content in carposporelings

Table 5 shows the pigment content of carpospore seedlings after 35 days of cultivation. Cultivated under conditions of 30°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity, treatment group 13 had the highest levels of Chla (0.36 $\text{mg}\cdot\text{g}^{-1}$) and Car (0.22 $\text{mg}\cdot\text{g}^{-1}$). In contrast, treatment group 8, grown at 20°C, 160 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 20 salinity, exhibited the lowest pigment levels, with Chla at 0.03 $\text{mg}\cdot\text{g}^{-1}$ and Car at 0.07 $\text{mg}\cdot\text{g}^{-1}$. Chla and Car levels decreased with increasing light intensity, reaching their highest levels (0.32 $\text{mg}\cdot\text{g}^{-1}$ and 0.21 $\text{mg}\cdot\text{g}^{-1}$, respectively) at 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$. Conversely, pigment levels increased with rising temperature. Chla content increased steadily with salinity, whereas Car content rose initially before declining. Variance analysis indicated that light intensity significantly affected Chla levels ($df = 3$, $N = 16$, $F = 10.42$,

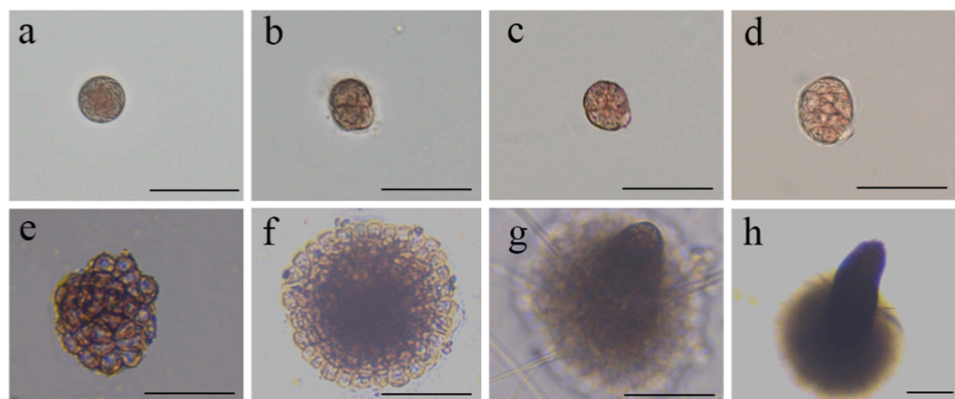


Fig. 1. Development of *Gracilaria fisheri* carpospores: a) freshly released carpospores; b) initial cell division; c) second cell division; d) carpospores post-multiple divisions; e) early disk; f) multicellular disk; g) disk center protrusion and filament formation; h) emergence of the erect thallus. scale bar = 50 μm .

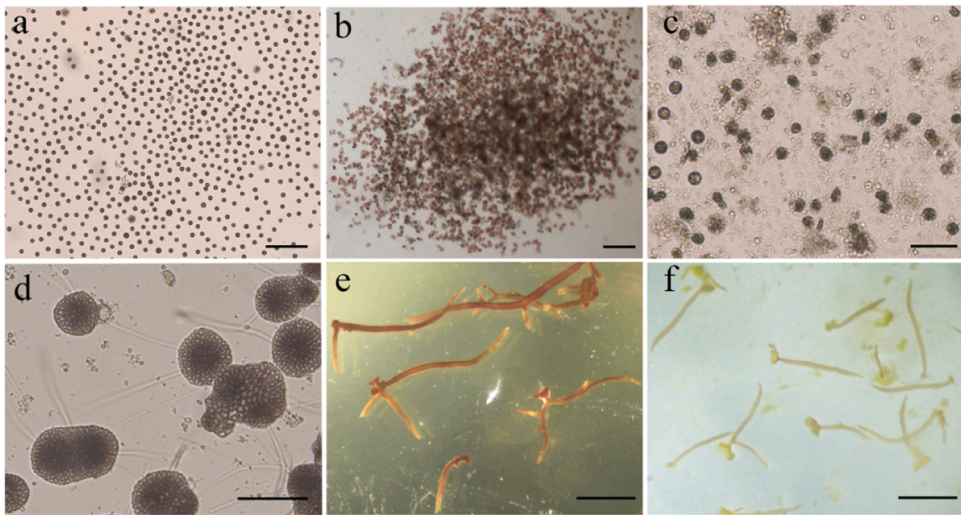


Fig. 2. Other phenomena that occur during the early development of carpospores: a) freshly released carpospores; b) a large aggregation of carpospores; c) ruptured carpospores; d) fusion phenomenon in multicellular disks; e) red carposporeling; f) yellow-green carposporeling. a-b scale bar = 100 μm , c-d scale bar = 50 μm , e-f scale bar = 1 mm.

Table 4
Orthogonal experiment and calculated results of linear growth rate of seedlings.

Group	Factor				Linear growth rate of seedlings (mm/day)
	A	B	C	Blank column	
1	15	20	20	1	−0.02
2	15	40	25	2	0.01
3	15	80	30	3	0.02
4	15	160	35	4	0.01
5	20	20	25	1	0.01
6	20	40	30	2	0.05
7	20	80	35	3	0.04
8	20	160	20	4	0.02
9	25	20	35	2	0.06
10	25	40	20	3	0.03
11	25	80	25	4	0.04
12	25	160	30	1	0.06
13	30	20	30	4	0.06
14	30	40	35	1	0.07
15	30	80	20	2	0.03
16	30	160	25	3	0.02
The range analysis of orthogonal experiment results for linear growth rate					
k1	0.000	0.003	0.002	0.03	
k2	0.03	0.04	0.02	0.03	
k3	0.05	0.03	0.05	0.03	
k4	0.04	0.03	0.04	0.03	
R	0.04	0.01	0.03	0.00	

k1, k2, k3 and k4 were the average values of Level 1, Level 2, Level 3 and Level 4 for each factor.
The range (R) was the difference between the maximum and minimum average ($k_{\text{max}} - k_{\text{min}}$) for each factor.
 k_{max} and k_{min} were the maximum and the minimum value of k1, k2, k3 and k4, respectively.

$P = 0.04$), while temperature and salinity had no significant effects ($df = 3, N = 16, F = 6.73, P = 0.08$; $df = 3, N = 16, F = 8.40, P = 0.06$). Variance analysis also indicated that light intensity significantly affected Car levels ($df = 3, N = 16, F = 12.15, P = 0.03$), while temperature and salinity had no significant effects ($df = 3, N = 16, F = 6.22, P = 0.08$; $df = 3, N = 16, F = 7.50, P = 0.07$). Based on the magnitude of variation in pigment levels (Table 5), the factors influencing Chla content were ranked as follows: light intensity ($R = 0.22$) > salinity ($R = 0.17$) = temperature ($R = 0.17$). For Car, the ranking was light intensity ($R = 0.09$) > temperature ($R = 0.07$) > salinity ($R = 0.06$). From the F-values of the variance analysis, the ranking of factors affecting Chla content

was refined to: light intensity ($F = 10.42$) > salinity ($F = 8.40$) > temperature ($F = 6.73$).

The phycobiliprotein content of carpospore seedlings after 35 days of cultivation is summarized in Table 5. Treatment group 9, grown under 25°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 35 salinity, exhibited the highest PE content (0.62 $\text{mg}\cdot\text{g}^{-1}$), and treatment group 13, grown under 30°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity, had the highest PC content (0.24 $\text{mg}\cdot\text{g}^{-1}$). In contrast, treatment group 16, grown at 30°C, 160 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 25 salinity, had the lowest levels of both pigments, with concentrations below 0.01 $\text{mg}\cdot\text{g}^{-1}$. Both phycobiliproteins increased initially and then declined with rising temperature, light intensity, and salinity. Variance analysis indicated that light intensity had no significant effect on PE content ($df = 3, N = 16, F = 1.18, P = 0.48$). The MS of salinity (0.45) and temperature (0.41) were smaller than the MSe (0.46) and thus was included in the error term. Variance analysis also indicated that light intensity had no significant effect on PC content ($df = 3, N = 16, F = 1.18, P = 0.48$). The MS of temperature (0.05) was smaller than the MSe (0.07) and thus was included in the error term. Based on the magnitude of variation in both phycobiliproteins, the primary influencing factors were ranked as follows: light intensity > salinity > temperature, which is consistent with the variance analysis results.

As shown in Tables 4 and 5, the optimal conditions for carpospore growth did not entirely align with those for pigment accumulation, and the best conditions for different pigments varied. Based on the k_{max} values shown in Table 5, the optimal conditions for pigment accumulation in carpospore seedlings were as follows: Chla: 30°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, 35 salinity; Car: 30°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, 30 salinity; PE: 25°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, 30 salinity; PC: 25°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, 30 salinity. Overall, the conditions of 25–30°C, 20–40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and a salinity range of 30–35 were suitable for the accumulation of all four pigments.

4. Discussion

Red algae have a complex and extended life cycle, and under natural conditions completion of a single life stage may span multiple seasons (Friedlander and Dawes, 1984; Gabrielson et al., 1986). The adaptive responses of red algae to environmental factors vary across different life stages (Guillemin et al., 2013). Temperature, light intensity, and salinity are fundamental factors that affect red algae growth and development (Dawes et al., 1998; Yokoya et al., 1999). Research on optimizing these factors has been central to advancing cultivation and seedling

Table 5
Orthogonal experiment and calculated results of the pigment content in seedlings.

Group	Factor				W _{Chla} (mg/g)	W _{Car} (mg/g)	W _{PE} (mg/g)	W _{PC} (mg/g)
	A	B	C	Blank column				
1	15	20	20	1	0.17	0.14	0.17	0.06
2	15	40	25	2	0.06	0.07	0.24	0.10
3	15	80	30	3	0.08	0.07	0.25	0.13
4	15	160	35	4	0.05	0.07	0.16	0.05
5	20	20	25	1	0.11	0.12	0.23	0.07
6	20	40	30	2	0.23	0.14	0.91	0.28
7	20	80	35	3	0.26	0.16	0.05	0.10
8	20	160	20	4	0.03	0.07	0.07	0.03
9	25	20	35	2	0.30	0.17	0.66	0.21
10	25	40	20	3	0.16	0.10	0.39	0.14
11	25	80	25	4	0.17	0.09	0.45	0.17
12	25	160	30	1	0.15	0.13	0.14	0.06
13	30	20	30	4	0.36	0.22	0.61	0.24
14	30	40	35	1	0.30	0.13	0.52	0.19
15	30	80	20	2	0.14	0.10	0.22	0.07
16	30	160	25	3	0.06	0.10	0.01	0.00
The range analysis of orthogonal experiment results for W _{Chla}								
k1	0.12	0.32	0.17	0.24				
k2	0.21	0.25	0.13	0.24				
k3	0.26	0.21	0.27	0.19				
k4	0.29	0.10	0.30	0.21				
R	0.17	0.22	0.17	0.06				
The range analysis of orthogonal experiment results for W _{Car}								
k1	0.12	0.21	0.14	0.17				
k2	0.16	0.15	0.13	0.16				
k3	0.16	0.14	0.19	0.14				
k4	0.18	0.13	0.17	0.15				
R	0.07	0.09	0.06	0.03				
The range analysis of orthogonal experiment results for W _{PE}								
k1	0.21	0.42	0.21	0.27				
k2	0.32	0.52	0.23	0.51				
k3	0.41	0.24	0.48	0.18				
k4	0.34	0.10	0.35	0.32				
R	0.21	0.42	0.27	0.33				
The range analysis of orthogonal experiment results for W _{PC}								
k1	0.08	0.14	0.08	0.10				
k2	0.12	0.18	0.09	0.17				
k3	0.15	0.12	0.18	0.09				
k4	0.13	0.04	0.14	0.12				
R	0.06	0.14	0.10	0.07				

k1, k2, k3 and k4 were the average values of Level 1, Level 2, Level 3 and Level 4 for each factor.
The range (R) was the difference between the maximum and minimum average ($k_{\max}-k_{\min}$) for each factor.
 $k_{\max}$ and $k_{\min}$ were the maximum and the minimum value of k1, k2, k3 and k4, respectively.

production of red algae in the family Gracilariaceae (Ramlov et al., 2012; Xu et al., 2009; Yu et al., 2013). In this study, we investigated the effects of temperature, light, and salinity on spore release, attachment, and germination and on seedling growth in *G. fisheri* using an orthogonal experimental design in order to identify optimal conditions for each developmental stage. Compared with the conventional approach of broadly collecting and cultivating spores, our study provides a more continuous and precise analysis of the early developmental stages of *G. fisheri* to clarify the influence of key environmental parameters on seedling establishment.

4.1. Influence of temperature, light intensity, and salinity on carpospore release

Spore release is the process whereby mature diploid spores are discharged from cystocarps on the female gametophyte into the external environment, and it is influenced by key environmental factors (Kain and Destombe, 1995). In our study, temperature and salinity significantly influenced spore release by *G. fisheri* ($P < 0.05$), with optimal release observed at 25°C, 30 salinity, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$. Temperature was identified as the primary factor affecting spore release in *G. fisheri*. Variations in optimal spore release temperatures among different *Gracilaria* species may reflect the temperature ranges typical of their native habitats. For example, Orduña-Rojas and Robledo (1999) found that the

optimal spore release temperature for *G. cornea*, a tropical species, was also 25°C. In contrast, Michetti et al. (2013) reported that the subtropical and temperate species *G. gracilis* released spores most effectively at 13–18°C. Salinity had the second most significant effect on spore release by *G. gracilis*, with an optimal 30 salinity. Azanza and Aliaza (1999) reported a similar optimal salinity for *Kappaphycus alvarezii* (30–35), but the value for both was higher than the range observed for *Catenella impudica* (19–29) (Narasimha Rao et al., 2008). Friedlander and Dawes (1984) reported that *G. foliifera* released the most spores at 25.2°C, 30 salinity, 80 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, which are conditions similar to those identified as optimal in our study. They also observed that under low-light or dark conditions, the optimal spore release temperature for *G. foliifera* dropped to 20°C; however, spore release in *G. fisheri* remained low at 20°C across all tested light intensities. While light intensity did not significantly impact *G. fisheri* spore release, the highest release was observed at the low light intensity of 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$. The enhancing effect of low light on spore release in red algae has been noted previously for *G. verrucosa* (= *G. gracilis*) (Sik et al., 2001) and *G. cornea* (Orduña-Rojas and Robledo, 1999).

In this study, the maximum daily release of *G. fisheri* carpospores reached 18,900 per gram fresh weight per day, with an estimated 280 carpospores released per cystocarp per day. This rate is comparable to that of *G. corticata*, which released 200–250 carpospores per cystocarp per day (Oza, 1975; Subba Rangaiah, 1983), and it is within the

mid-range of release rates reported for *G. foliifera* (1015–26,368 carpospores per gram per day) (Friedlander and Dawes, 1984). However, the release rate for *G. fisheri* was lower than those for *G. dura* and *G. gracilis*, which released 808 and 1194 carpospores per cystocarp per day, respectively (Mantri et al., 2009; Michetti et al., 2013). Researchers previously reported that carpospore release in *Gracilariopsis lemaneiformis* and *G. dura* was generally concentrated within a 7-day period, peaking on day 4 (Mantri et al., 2009; Zhou et al., 2016). In contrast, *G. verrucosa* (= *G. gracilis*) exhibited a cyclical release pattern, releasing carpospores every 4–5 days over the span of a month (Lefebvre et al., 1987; Oza and Krishnamurthy, 1968). In our study, *G. fisheri* did not exhibit a concentrated release pattern within a 7-day period; instead, carpospore release persisted for up to 1 month into the experiment. Given the relatively low average daily release rate compared to other *Gracilaria* species, *G. fisheri* may be characterized by a long-term cyclical release pattern.

4.2. Influence of temperature, light intensity, and salinity on carpospore attachment and germination

After release, *G. fisheri* carpospores settle onto the substrate, initiating cell division to form a central, thickened disk that anchors to the substrate surface. From the disk's center, an erect thallus emerges, which progresses to juvenile plant development (Santelices and Doty, 1989). *G. fisheri* carpospores generally follow the *Dumontia* type of germination pattern, in which a mass of cells forms before the production of an erect thallus (Dixon, 1973).

In our study, both temperature and salinity markedly influenced the attachment and germination rates of *G. fisheri* carpospores, with salinity being the dominant factor in both processes. The optimal salinity for *G. fisheri* carpospore attachment and germination was 30, which is comparable to the optimal salinity range of 28–30 reported for *G. edulis* and *G. tenuistipitata* spores (Yu et al., 2013). Lower salinity levels resulted in a dramatic decrease in attachment and germination rates. For example, at salinity of 15, many carpospores ruptured and died (Fig. 2c), with germination rates dropping below 1 %. Similar sensitivity to low salinity (10–20) has been observed in other red algae, such as *G. domingensis* and *Hidropuntia caudata* (de Miranda et al., 2012; Ramlov et al., 2012). The optimal temperature range for attachment and germination of red algal carpospores may reflect their adaptation to specific habitats. *G. fisheri* carpospores showed partial germination within a temperature range of 20°C to 30°C, while attachment and germination rates dropped to zero at 15°C. In contrast, Li et al. (2010) found that the red alga *Chondrus ocellatus*, which is native to higher latitudes, has carpospores that can survive at temperatures between 10 and 25°C. Although it did not significantly impact germination rates, light intensity significantly influenced attachment rates. At 160 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, the attachment rate was only 13 %. This result may suggest that the multicellular structure of disks enhances their tolerance to high light levels.

Interestingly, we also observed a tendency for neighboring *G. fisheri* carpospores to merge during disk formation, resulting in fused disks with two or more carpospores (Fig. 2d). Such fusion likely strengthens attachment to the substrate and increases the survival rate of juvenile plants (Santelices et al., 1996; Yu et al., 2016). Prior studies indicate that not all fused disks develop erect thalli; germination may depend on the disk's diameter, and smaller multicellular disks formed by fusion may lack the ability to produce shoots (Alveal et al., 1997; Santelices, 2004). This tendency could partly explain the difference in attachment and germination rates observed in *G. fisheri*.

4.3. Influence of temperature, light intensity, and salinity on the growth and pigment content of carposporelings

Carposporelings eventually release tetraspores as they mature. These tetraspores develop into male and female gametophytes, marking the

progression to the next stage in the life cycle. We found that the *G. fisheri* carposporeling linear growth rate was significantly affected by temperature and salinity, with temperature being the primary influencing factor. In this study, the optimal growth conditions for *G. fisheri* carposporelings were 25°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity, which are very similar to those reported for *Gp. lemaneiformis* carpospore development (25°C, 45 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, 30) (Zhou et al., 2013). Likewise, Yu et al. (2013) found that *G. edulis* and *G. tenuistipitata* carposporelings exhibited the highest linear growth rates at $25 \pm 2^\circ\text{C}$, although they thrived at a lower optimal salinity of 15. Variations in salinity preferences may correlate with the salinity tolerance of adult *Gracilaria*. *G. edulis* and *G. tenuistipitata*, for instance, are known euryhaline species (Rebello et al., 1996; Xu et al., 2009), with *G. tenuistipitata* adults growing most rapidly under low-salinity conditions (20) (Yokoya et al., 1999).

Under optimal conditions, *G. fisheri* carpospores can develop into visible thalli within 4 weeks, which is earlier than the 5 weeks required for *Gp. lemaneiformis* but later than the 3 weeks reported for *G. cornea* (Orduña-Rojas and Robledo, 1999; Zhou et al., 2013). Glenn et al. (1996) reported that carposporelings of *G. parvispora* require 18–20 weeks to grow to a size of 0.02–0.6 cm. In contrast, the carposporelings of *G. fisheri* in this study reached approximately 0.5 cm in just 9 weeks.

The study revealed that a light intensity of 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ was optimal for *G. fisheri* carposporeling growth, but it was lower than values reported in some other *Gracilaria* species (Bunsom and Prathep, 2012; Xu et al., 2009; Yu et al., 2013). This difference could be due to the use of mature, branched *Gracilaria* specimens in these other studies. We used young *G. fisheri* carposporelings, which may be more sensitive to photoinhibition due to a higher surface-area-to-volume ratio or limited photoprotective mechanisms. The study also revealed that light intensity significantly influenced the accumulation of Chla and Car in carposporelings. With increasing light intensity, *G. fisheri* carposporeling Chla content gradually declined. This negative correlation was also observed in *G. tenuistipitata* and *G. verrucosa* (= *Gp. longissima*) (Ak and Yücesan, 2012; Carnicas et al., 1999). Aside from acting as light-harvesting pigments, Car also serve as photoprotective agents by shielding the photosystem under high light via the xanthophyll cycle (Rmiki et al., 1996). Torres et al. (2015) studied the effect of high light intensity on Car content in *Gp. tenuifrons* and found that β -carotene content was reduced by 42 % and 55 % at 600 and 1000 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, respectively, compared to the control (60 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$). Torres et al. also found that zeaxanthin content initially increased before declining, and after 7 days it remained below the control level. Although the light intensity in our study was much lower than that tested by Torres et al. (2015), *G. fisheri* Car content followed a similar trend, decreasing as light intensity increased, with a 38 % reduction from 20 to 160 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$. We only measured Car content at day 35, so it is unclear if Car levels increased in the short term, and this possibility warrants further research. Ak and Yücesan (2012) reported that PE and PC, which are auxiliary light-harvesting pigments, also decreased in *Gp. longissima* with increasing light intensity. In *G. fisheri*, however, this trend only became apparent when light intensity exceeded 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$. Bermejo et al. (2020) reported that PE content in *Gp. longissima* was lower under low light (60 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) than under high light, with nutrient-rich conditions amplifying this difference. This may explain why light intensity had no significant effect on PE content in our study ($P > 0.05$). Pigment levels were measured on day 35, 7 days after the last medium change, suggesting that nutrient depletion may have reduced PE content differences across treatments.

Long-term exposure to varying light intensities can cause changes in the pigment content and composition of seaweeds (Machalek et al., 1996) and affect thallus color on a macroscopic scale (Torres et al., 2015). We found that *G. fisheri* carposporelings exhibited red and yellow-green coloration (Fig. 2e, f). Under 160 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ light intensity, the thalli appeared uniformly yellow-green, and Chla content in groups 4, 8, and 16 was the lowest of all treatments. Group 16, which was cultured at 30°C, had the lightest coloration, the slowest growth,

and the lowest levels of PE and PC among all groups. Several researchers have reported that red algae can appear yellow or brown during high-light summer conditions and red-brown during low-light winter conditions (Calabrese, 1972; Waaland et al., 1974). The culture conditions in group 16, which mimicked high-light, high-temperature summer conditions, likely influenced these results. High temperatures can reduce the positive effects of nitrogen on thalli by limiting the synthesis of stress-protective proteins, thereby intensifying the impact of high light (Hurd et al., 2014; Ma et al., 2021).

Using spores as seed material can effectively address issues such as population degradation and yield reduction caused by excessive vegetative propagation. It ensures a continuous supply of healthy and uniform *Gracilaria* seedlings, which is of great significance for the future large-scale cultivation of economically valuable red seaweeds (Buschmann and Kuschel, 1988). In this study, we focused on *G. fisheri* and investigated how temperature, light intensity, and salinity influence carposporeling growth. By addressing the gap in carposporeling research for this species, we identified optimal conditions for carpospore dispersal, attachment, and germination and for seedling growth. These optimal conditions could mitigate challenges in spore-based cultivation, such as lengthy seedling cycles and low seedling success rates, while also eliminating individuals with weaker tolerance to cultivation environments. This would provide healthier seedlings for expanded cultivation and marine transplantation, which in turn would support factory-scale seedling propagation and future artificial breeding research in *Gracilaria*.

5. Conclusion

In this study, we successfully obtained *G. fisheri* carpospores and cultivated them into seedlings. Using a three-factor, four-level orthogonal experiment, we identified the optimal conditions for various stages of *G. fisheri* development. The best conditions for carpospore release were 25°C, 20 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity. For carpospore attachment, the optimal conditions were 20°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity. For carpospore germination, 30°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity were ideal. The best conditions for carposporeling growth were 25°C, 40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30 salinity. Under suitable conditions, visible carposporeling of *G. fisheri* could be observed within 4 weeks, and the maximum length reached 0.79 cm after 9 weeks. Suitable conditions for pigment accumulation in carposporelings were 25–30°C, 20–40 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and 30–35 salinity.

CRediT authorship contribution statement

Wang Huihui: Methodology. **Xin Rong:** Investigation. **Lin Kun:** Writing – review & editing, Software. **Zeng Zhaojun:** Methodology, Investigation. **Guo Youyou:** Methodology, Investigation. **Chen Chunli:** Resources. **He Haiping:** Investigation. **Xie Enyi:** Supervision, Project administration, Methodology. **Cui Jianjun:** Writing – review & editing, Funding acquisition. **Yang Wencheng:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Wu Jinhui:** Investigation. **Liao Jiawei:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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